

Problem 15.20

Seth, Janice, and Lori each borrow \$5,000 for five years at a nominal interest rate of 12%, compounded semiannually. Seth has interest accumulated over the five years and pays all the interest and principal in a lump sum at the end of five years. Janice pays interest at the end of every six-month period as it accrues and the principal at the end of five years. Lori repays her loan with 10 level payments at the end of every six-month period. Calculate the total amount of interest paid on all three loans.

Solution

The nominal annual interest rate is 12% compounded semiannually, so the six-month effective interest rate is

$$j = \frac{0.12}{2} = 0.06.$$

Since the term is five years, there are

$$5(2) = 10$$

six-month periods.

Seth

Seth makes one lump-sum payment at the end of five years. The accumulated value of the loan is

$$5000(1.06)^{10}.$$

Thus, Seth's interest paid is

$$5000(1.06)^{10} - 5000 \approx 3954.2385.$$

Janice

Janice pays the interest as it accrues every six months. Each interest payment is

$$5000(0.06) = 300.$$

Since there are 10 six-month periods, Janice's total interest paid is

$$10(300) = 3000.$$

Lori

Lori repays the loan with 10 level semiannual payments. If each payment is X , then

$$5000 = Xa_{\overline{10}|}$$

Using

$$a_{\overline{n}|} = \frac{1 - (1 + j)^{-n}}{j},$$

we get

$$\begin{aligned} X &= \frac{5000}{a_{\overline{10}|}} \\ &= \frac{5000}{\frac{1 - (1.06)^{-10}}{0.06}} \\ &\approx 679.3398. \end{aligned}$$

Lori's total payments are $10X$, so her total interest paid is

$$10(679.3398) - 5000 \approx 1793.3979.$$

Therefore, the total interest paid on all three loans is

$$3954.2385 + 3000 + 1793.3979 \approx 8747.6364.$$

Hence, the total amount of interest paid is

$$\boxed{\$8,747.64}.$$